# Proof for Remark 7.22

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## Problem Statement

The main goal of this task is to investigate Remark 7.22; the rest is background. We do not know if it holds for $d \geqslant 3$, and this is the only question.

**Example 7.21.** Let $R$ be a commutative ring, and let $x \in R$ be a nonzerodivisor. For integers $a, b, c \geqslant 0$, define a complex

$$X(a,b,c) = \bigoplus_{i \geqslant 0} (R/f_i)[i] = \left( \cdots \xrightarrow{0} R/f_2 \xrightarrow{0} R/f_1 \xrightarrow{0} R/f_0 \to 0 \right),$$

where $f_i = x^{ai^2+bi+c} \in R$. Then

$$\operatorname{thick}^{\otimes}\{X(a,b,c) \mid a,b,c \geqslant 0\} = \operatorname{thick}^{\otimes}\{X(1,0,0)\}.$$

**Remark 7.22.** One can consider a general statement of Example 7.21 by defining

$$f_i = x^{a_0 i^d + a_1 i^{d-1} + \cdots + a_d},$$

so that Example 7.21 is exactly the case $d = 2$. We do not know if it holds for $d \geqslant 3$.

## Proof

We prove that the general statement holds for all integers $d \geqslant 1$. Specifically, for any integer $d \geqslant 1$, any commutative noetherian ring $R$, and any nonzerodivisor $x \in R$, define

$$X(a_0, a_1, \ldots, a_d) = \bigoplus_{i \geqslant 0} \left( R/x^{a_0 i^d + a_1 i^{d-1} + \cdots + a_d} \right)[i]$$

for non-negative integers $a_0, \ldots, a_d$. We claim that

$$\operatorname{thick}^{\otimes}\{X(a_0, a_1, \ldots, a_d) \mid a_0, \ldots, a_d \geqslant 0\} = \operatorname{thick}^{\otimes}\{X(1, 0, \ldots, 0)\}.$$

The inclusion ($\supseteq$) is trivial since $X(1,0,\ldots,0)$ appears on the left-hand side with $a_0 = 1$ and $a_1 = \cdots = a_d = 0$. For the inclusion ($\subseteq$), we must show that every $X(a_0, \ldots, a_d)$ belongs to $\operatorname{thick}^{\otimes}\{X(1,0,\ldots,0)\}$. We abbreviate

$$\mathcal{T} = \operatorname{thick}^{\otimes}\{X(1,0,\ldots,0)\}$$

throughout.

Let $e_k$ denote the $(d+1)$-tuple with 1 in position $k$ and 0 elsewhere, for $0 \leqslant k \leqslant d$. By Lemma 7.20 (second part, iterated), the complex $X(a_0, \ldots, a_d)$ whose exponent at index $i$ is $a_0 i^d + a_1 i^{d-1} + \cdots + a_d$ belongs to the thick closure of the complexes $X(e_k)$ for $k = 0, \ldots, d$. Since $\operatorname{thick}^{\otimes}$ is at least as large as the thick closure, it suffices to show $X(e_k) \in \mathcal{T}$ for each $k$.

Now $X(e_k)$ has exponent sequence $\{i^{d-k}\}_{i\geqslant 0}$. Define

$$\mathbf{X}(p) = \bigoplus_{i\geqslant 0}(R/x^{p(i)})[i]$$

for any sequence $p(i) \geqslant 0$. Then our goal reduces to showing that for each $0 \leqslant k \leqslant d$, the complex $\mathbf{X}(i^{d-k})$ belongs to $\mathcal{T} = \mathrm{thick}^{\otimes}\{\mathbf{X}(i^d)\}$. Equivalently, for each $0 \leqslant m \leqslant d$, we must show $\mathbf{X}(i^m) \in \mathcal{T}$.

The proof requires two new ingredients beyond those used in Example 7.21: a coefficient formula for finite differences, and a generalization of Proposition 7.2 / Corollary 7.3 from 2-fold to $p$-fold splitting. We establish these first.

## Lemma A (Coefficient Formula for Finite Differences)

**Lemma 1.** *For integers $d \geqslant 1$ and $0 \leqslant m \leqslant d$, the $m$-th forward finite difference of $i^d$ has the monomial expansion*

$$\Delta^m(i^d) = \sum_{j=0}^{d-m}\binom{d}{j}\, m!\, S(d-j,m)\, i^j,$$

*where $S(n,r)$ denotes the Stirling number of the second kind. In particular:*

1. *Every coefficient $\binom{d}{j}\, m!\, S(d-j,m)$ is a non-negative integer.*
2. *The leading coefficient, at $j = d-m$, equals $\dfrac{d!}{(d-m)!}$.*
3. *$\Delta^m(i^d) \geqslant 0$ for all $i \geqslant 0$.*

*Proof.* The $m$-th forward finite difference is

$$\Delta^m(i^d) = \sum_{s=0}^{m}(-1)^{m-s}\binom{m}{s}(i+s)^d.$$

Since the coefficient of $i^j$ in $(i+s)^d$ is $\binom{d}{j}s^{d-j}$, the coefficient of $i^j$ in $\Delta^m(i^d)$ is

$$\binom{d}{j}\sum_{s=0}^{m}(-1)^{m-s}\binom{m}{s}s^{d-j}.$$

By the standard identity for Stirling numbers of the second kind,

$$\sum_{s=0}^{m}(-1)^{m-s}\binom{m}{s}s^n = m!\, S(n,m)$$

for all $n \geqslant 0$, with the convention $S(0,0) = 1$ and $S(n,m) = 0$ for $n < m$. Therefore the coefficient of $i^j$ is exactly $\binom{d}{j}\, m!\, S(d-j,m)$.

For (1), each factor $\binom{d}{j}$, $m!$, and $S(d-j,m)$ is a non-negative integer. For (2), at $j = d-m$, the coefficient is

$$\binom{d}{d-m}m!S(m,m) = \frac{d!}{(d-m)!\, m!}\cdot m!\cdot 1 = \frac{d!}{(d-m)!}.$$

For $j > d-m$, we have $d-j < m$, so $S(d-j,m) = 0$; thus the polynomial has degree exactly $d-m$. For (3), since

$$\Delta^m(i^d) = \sum_j \alpha_j i^j$$

with all $\alpha_j \geqslant 0$, and $i \geqslant 0$, we get $\Delta^m(i^d) \geqslant 0$. $\square$

### Lemma B ($p$-fold Splitting, Generalized Corollary 7.3)

**Lemma 2.** *Let $p \geqslant 2$ be an integer, and let*

$$X = \bigoplus_{n \geqslant 0} M_n[n]$$

*be a complex in $\mathrm{D}^-(R)$ with zero differentials. For $r = 0, 1, \ldots, p-1$, define*

$$X^{(r)} = \bigoplus_{i \geqslant 0} M_{pi+r}[i].$$

*Then*

$$X \in \operatorname{thick}^{\otimes}\{X^{(0)}, X^{(1)}, \ldots, X^{(p-1)}\}.$$

*Proof.* We show that for each $r \in \{0, \ldots, p-1\}$, the complex

$$Z_r := \bigoplus_{i \geqslant 0} M_{pi+r}[pi + r]$$

is in $\operatorname{thick}^{\otimes}\{X^{(r)}\}$. Since

$$X = Z_0 \oplus Z_1 \oplus \cdots \oplus Z_{p-1}$$

as a direct sum decomposition of graded modules with zero differentials, the conclusion follows.

Fix $r \in \{0, \ldots, p-1\}$. Define the complex

$$Y_r = \bigoplus_{j \geqslant 0} R[(p-1)j + r].$$

Under the convention that $[k]$ places the module in homological degree $k$ (and thus cohomological degree $-k$), $Y_r$ has non-zero components only in cohomological degrees $\leqslant 0$, so $Y_r \in \mathrm{D}^-(R)$.

Because $X^{(r)}$ is a complex with zero differentials and $Y_r$ is a complex of free $R$-modules, hence $K$-flat, the derived tensor product $X^{(r)} \otimes^L_R Y_r$ equals the ordinary tensor product and has zero differentials. Explicitly,

$$X^{(r)} \otimes^L_R Y_r = \bigoplus_{i,j \geqslant 0} (M_{pi+r} \otimes_R R)[i + (p-1)j + r] = \bigoplus_{i,j \geqslant 0} M_{pi+r}[i + (p-1)j + r].$$

In homological degree $n$, this tensor product has module

$$\bigoplus_{i+(p-1)j+r=n} M_{pi+r}.$$

Consider the diagonal subcollection defined by $j = i$. For $j = i$, the degree is

$$i + (p-1)i + r = pi + r,$$

and the module contributed is $M_{pi+r}$. Thus, for each $i_0 \geqslant 0$, the module $M_{pi_0+r}$ appears as a direct summand of the degree-$(pi_0 + r)$ component of $X^{(r)} \otimes^L_R Y_r$.

Since all differentials are zero, the subcomplex

$$Z_r = \bigoplus_{i \geqslant 0} M_{pi+r}[pi + r]$$

is a direct summand of $X^{(r)} \otimes^L_R Y_r$ in the category of complexes, hence in $\mathrm{D}^-(R)$. Since thick tensor ideals are closed under tensor product with any object of $\mathrm{D}^-(R)$, we have

$$X^{(r)} \otimes^L_R Y_r \in \operatorname{thick}^{\otimes}\{X^{(r)}\}.$$

Because thick tensor ideals are closed under direct summands, we conclude $Z_r \in \operatorname{thick}^{\otimes}\{X^{(r)}\}$.

Therefore

$$X = \bigoplus_{r=0}^{p-1} Z_r \in \operatorname{thick}^{\otimes}\{X^{(0)}, X^{(1)}, \ldots, X^{(p-1)}\}.$$

$\square$

## Lemma C (Subtraction of Exponent Sequences)

**Lemma 3.** *If the complexes $\mathbf{X}(a_i + b_i)$ and $\mathbf{X}(b_i)$ both belong to $\mathcal{T}$, and $a_i, b_i \geqslant 0$ for all $i \geqslant 0$, then $\mathbf{X}(a_i)$ also belongs to $\mathcal{T}$.*

*Proof.* For each $i \geqslant 0$, we have a short exact sequence of $R$-modules

$$0 \to R/x^{a_i} \xrightarrow{x^{b_i}} R/x^{a_i+b_i} \to R/x^{b_i} \to 0,$$

where the map is well-defined and injective since $x$ is a nonzerodivisor. Taking the direct sum over all $i \geqslant 0$ and placing each term in degree $[i]$, we obtain a short exact sequence of complexes with zero differentials:

$$0 \to \mathbf{X}(a_i) \to \mathbf{X}(a_i + b_i) \to \mathbf{X}(b_i) \to 0.$$

Since $\mathcal{T}$ is a thick subcategory, closed under extensions and cones, and both $\mathbf{X}(a_i + b_i)$ and $\mathbf{X}(b_i)$ are in $\mathcal{T}$, the two-out-of-three property gives $\mathbf{X}(a_i) \in \mathcal{T}$. □

## Lemma D (Finite Differences Preserve the Thick Tensor Ideal)

**Lemma 4.** *Let $\{a_i\}_{i\geqslant 0}$ be a sequence of non-negative integers such that $a_{i+1} \geqslant a_i$ for all $i \geqslant 0$. If $\mathbf{X}(a_i) \in \mathcal{T}$, then $\mathbf{X}(a_{i+1} - a_i) \in \mathcal{T}$.*

*Proof.* For each $i \geqslant 0$, the short exact sequence

$$0 \to R/x^{a_i} \xrightarrow{x^{a_{i+1}-a_i}} R/x^{a_{i+1}} \to R/x^{a_{i+1}-a_i} \to 0$$

yields, upon taking direct sums over $i$, a short exact sequence of complexes with zero differentials:

$$0 \to \mathbf{X}(a_i) \to \mathbf{X}(a_{i+1}) \to \mathbf{X}(a_{i+1} - a_i) \to 0.$$

By the two-out-of-three property for thick subcategories, it suffices to show that $\mathbf{X}(a_{i+1}) \in \mathcal{T}$.

By re-indexing $j = i + 1$, we have

$$\mathbf{X}(a_{i+1}) = \bigoplus_{i\geqslant 0}(R/x^{a_{i+1}})[i] = \bigoplus_{j\geqslant 1}(R/x^{a_j})[j-1] = \left(\bigoplus_{j\geqslant 1}(R/x^{a_j})[j]\right)[-1].$$

Since

$$\mathbf{X}(a_i) = \bigoplus_{j\geqslant 0}(R/x^{a_j})[j]$$

is a complex with zero differentials, it decomposes as a direct sum of its graded pieces, so

$$\bigoplus_{j\geqslant 1}(R/x^{a_j})[j]$$

is a direct summand of $\mathbf{X}(a_i) \in \mathcal{T}$, and hence belongs to $\mathcal{T}$. Applying the shift $[-1]$ gives $\mathbf{X}(a_{i+1}) \in \mathcal{T}$. □

**Corollary of Lemma D.** By iterating Lemma D, if $\mathbf{X}(a_i) \in \mathcal{T}$ and the sequence $\{a_i\}$ satisfies $\Delta^l(a_i) \geqslant 0$ for all $i \geqslant 0$ and $l = 1, \ldots, m$, then $\mathbf{X}(\Delta^m(a_i)) \in \mathcal{T}$. In particular, since $\Delta^l(i^d) \geqslant 0$ for all $i \geqslant 0$ and $0 \leqslant l \leqslant d$ by Lemma A, starting from $\mathbf{X}(i^d) \in \mathcal{T}$ we get $\mathbf{X}(\Delta^m(i^d)) \in \mathcal{T}$ for every $0 \leqslant m \leqslant d$.

## Proof of the Main Claim

We prove by induction on $k$, from $k = 0$ to $k = d$, that $\mathbf{X}(i^k) \in \mathcal{T}$.

**Base case ($k=0$).** The complex $\mathbf{X}(1)$ has exponent sequence $\{1,1,1,\ldots\}$. Since

$$\mathbf{X}(i^d) = \bigoplus_{i\geqslant 0}(R/x^{i^d})[i]$$

has $(R/x^{1^d})[1] = (R/x)[1]$ as a direct summand, the module $R/x$ belongs to $\mathcal{T}$. Then

$$\mathbf{X}(1) = (R/x) \otimes^L_R \left(\bigoplus_{i\geqslant 0} R[i]\right) \in \mathcal{T}$$

since $\bigoplus_{i\geqslant 0} R[i] \in \mathrm{D}^-(R)$ and thick tensor ideals are closed under tensor products.

**Inductive step.** Suppose $k \geqslant 1$ and $\mathbf{X}(i^l) \in \mathcal{T}$ for all $0 \leqslant l < k$. We prove $\mathbf{X}(i^k) \in \mathcal{T}$. Set $m = d-k$. By the corollary of Lemma D, the complex $\mathbf{X}(Q(i)) \in \mathcal{T}$, where

$$Q(i) = \Delta^m(i^d).$$

By Lemma A,

$$Q(i) = \sum_{j=0}^{k} \alpha_j i^j, \qquad \alpha_j = \binom{d}{j} m! S(d-j,m) \geqslant 0,$$

and

$$\alpha_k = \frac{d!}{k!}.$$

Define

$$c = \alpha_k = \frac{d!}{k!}, \qquad q(i) = Q(i) - c\, i^k = \sum_{j=0}^{k-1} \alpha_j i^j.$$

**Step 1.** We show $\mathbf{X}(q(i)) \in \mathcal{T}$. Since each $\alpha_j$ is a non-negative integer by Lemma A, and $\mathbf{X}(i^j) \in \mathcal{T}$ for $j < k$ by the inductive hypothesis, Lemma 7.20 (scalar multiplication) gives $\mathbf{X}(\alpha_j i^j) \in \mathcal{T}$. Summing with Lemma 7.20 (addition), we get $\mathbf{X}(q(i)) \in \mathcal{T}$.

**Step 2.** We show $\mathbf{X}(c\, i^k) \in \mathcal{T}$. Since $Q(i) = c\, i^k + q(i)$ with $c\, i^k \geqslant 0$ and $q(i) \geqslant 0$, Lemma C applied with $a_i = c\, i^k$ and $b_i = q(i)$ gives $\mathbf{X}(c\, i^k) \in \mathcal{T}$.

**Step 3.** We show $\mathbf{X}(i^k) \in \mathcal{T}$ via $c$-fold splitting. Apply Lemma B with $p = c$ to the complex $\mathbf{X}(i^k)$. The $c$-fold parts are

$$\mathbf{X}(i^k)^{(r)} = \mathbf{X}((ci+r)^k), \qquad r = 0,\ldots,c-1.$$

By Lemma B,

$$\mathbf{X}(i^k) \in \mathrm{thick}^{\otimes}\{\mathbf{X}((ci+r)^k) \mid r = 0,\ldots,c-1\}.$$

It remains to show each $\mathbf{X}((ci+r)^k) \in \mathcal{T}$.

By the binomial theorem,

$$(ci+r)^k = c^k i^k + \sum_{j=0}^{k-1} \binom{k}{j} c^j r^{k-j} i^j.$$

For the lower-order terms, each coefficient

$$\beta_j = \binom{k}{j} c^j r^{k-j}$$

is a non-negative integer. By the inductive hypothesis, $\mathbf{X}(i^j) \in \mathcal{T}$, so $\mathbf{X}(\beta_j i^j) \in \mathcal{T}$, and their sum is in $\mathcal{T}$. For the leading term, Step 2 gives $\mathbf{X}(c\,i^k) \in \mathcal{T}$. By Lemma 7.20 (scalar multiplication with factor $c^{k-1}$),

$$\mathbf{X}(c^k i^k) = \mathbf{X}(c^{k-1} \cdot c\,i^k) \in \mathcal{T}.$$

Adding the leading and lower-order terms using Lemma 7.20 (addition), we get $\mathbf{X}((ci+r)^k) \in \mathcal{T}$. By Lemma B, this implies $\mathbf{X}(i^k) \in \mathcal{T}$.

### Completing the Proof

By the main claim, $\mathbf{X}(i^m) = X(e_{d-m}) \in \mathcal{T}$ for all $0 \leqslant m \leqslant d$. By Lemma 7.20 (second part, iterated), for any tuple $(a_0, \dots, a_d)$ of non-negative integers, the complex $X(a_0, \dots, a_d)$ with exponent

$$\sum_j a_j i^{d-j}$$

belongs to the thick closure of $\{X(e_0), X(e_1), \dots, X(e_d)\}$, which is contained in $\mathcal{T}$. Therefore

$$\text{thick}^{\otimes}\{X(a_0, \dots, a_d) \mid a_0, \dots, a_d \geqslant 0\} \subseteq \mathcal{T} = \text{thick}^{\otimes}\{X(1, 0, \dots, 0)\},$$

proving the general statement.

## Key Ideas

1. **$p$-fold splitting (Lemma B).** This generalizes Corollary 7.3 to arbitrary $p \geqslant 2$, enabling division by $p$ for any $p$. It is proved by tensoring with $Y_r = \bigoplus_j R[(p-1)j+r]$ and extracting the diagonal direct summand.

2. **Coefficient formula via Stirling numbers (Lemma A).** The finite difference $\Delta^{d-k}(i^d)$ is a degree-$k$ polynomial whose lower-degree terms have non-negative integer coefficients, enabling subtraction through Lemma C.

3. **Bottom-up induction.** Starting from $\mathbf{X}(i^d) \in \mathcal{T}$, we iterate finite differences to produce lower-degree polynomials, subtract off strictly lower-degree terms already known to lie in $\mathcal{T}$, and use $c$-fold splitting to isolate $\mathbf{X}(i^k)$.